

MINGYU SEONG

Wearable Robotics | Lower-Limb Exoskeletons | Rehabilitation Robotics | Human-Robot Interaction

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EDUCATION

Yeungnam University, Department of Robotics Engineering

Feb. 2026

M.S. in Robotics Engineering | GPA: 4.37 / 4.50

Thesis: "Development of Personalized Robotic Achilles Tendon based on Switchable-Stiffness Spring"

Advisor: Prof. Jungsu Choi

Yeungnam University, Department of Robotics Engineering

Feb. 2024

B.Eng. in Robotics Engineering | GPA: 4.20 / 4.50

RESEARCH EXPERIENCE

Human Robot Convergence Laboratory (HRC Lab), Yeungnam University

Gyeongsan, Korea

Researcher | Mar. 2021 – Present | Advisor: Prof. Jungsu Choi

- Developed a 0.99-kg semi-passive ankle exoskeleton using switchable-stiffness pneumatic springs (M.S. thesis; patent pending).
- Led the full development cycle: mechanism design, sensing and gait-phase control, and human-subject testing.
- Reduced the metabolic cost of walking by 14.3% compared with walking without the device.

PUBLICATIONS

Journal Article (Under Review)

M. Seong and J. Choi*, "Robotic Achilles Tendon: A Lightweight Semi-Passive Ankle-Assist System for Reducing the Metabolic Cost of Walking," *IEEE Robotics & Automation Magazine*, under review.

Peer-Reviewed International Conference Papers

M. Seong, H. Heo, H. Lee, and J. Choi*, "Personalized Robotic Achilles Tendon Utilizing a Semi-Passive Spring with Switching Stiffness," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. DOI: 10.1109/IROS60139.2025.11245843

M. Seong, H. Heo, H. Lee, and J. Choi*, "A Novel Wearable Ankle Assistive System Utilizing Switching Stiffness Pneumatic Spring," *International Conference on Control, Automation, and Systems (ICCAS)*, 2025. DOI: 10.23919/ICCAS66577.2025.11301378

Domestic Conference Papers (in Korean, with English abstracts)

M. Seong, H. Heo, and J. Choi*, "Development of Bioinspired Variable Stiffness Actuating System based on Human Musculoskeletal Characteristics," *ICROS Annual Conference*, 2025.

M. Seong, H. Heo, and J. Choi*, "Development of Semi-passive Ankle Assist System Based on Lower Limb Dynamic Characteristic," *Korea Robotics Society Annual Conference (KRoC)*, 2025.

M. Seong and J. Choi*, "Development of Bioinspired Force Transmission Structure based on Human Sarcomere by using Fluid Dynamics," *ICROS Annual Conference*, 2024. — Outstanding Paper Award

M. Seong, Y. Lee, and J. Choi*, "New Measurement Approach for Dynamic Force Utilizing Fluid," *Korea Robotics Society Annual Conference (KRoC)*, 2023.

* Corresponding author

CONFERENCE PRESENTATIONS

M. Seong, H. Heo, and J. Choi*, "Semi-passive Ankle Assist System Based on Lower Limb Dynamic Characteristics of Human Walking,"

Oral Presentation, *Joint 10th IFAC Symposium on Mechatronic Systems and 14th IFAC Symposium on Robotics*, Paris, France, July 2025.

PATENTS

J. Choi and **M. Seong**, "Ankle Assist Device," Korean Patent Application No. 10-2026-0011170, filed Jan. 20, 2026 (pending).

SELECTED FUNDED RESEARCH PROJECTS (PARTICIPATING RESEARCHER)

Inner-Wear-Type Wearable Parasitic Muscle Suit to Overcome the Limitations of Human Physical Ability

Mar. 2023 – Feb. 2028

National Research Foundation of Korea (NRF) — Outstanding Young Scientist Grants

HD-DNA Lab: Human Behavior Stem Information-Based Digital Human Cloning Innovative Supplementary Engineering Laboratory

Jun. 2022 – Feb. 2025

National Research Foundation of Korea (NRF) — Basic Research Laboratory (Pioneer)

Develop of Exomuscle Device to Improve a Gait Disturbance for People with Depressed Ankle Motion

Jun. 2021 – Feb. 2023

National Research Foundation of Korea (NRF) — Basic Research Grant

Development of High-Speed Winding Equipment for Tow Prepreg-Based Hydrogen Storage Vessels

Jun. 2021 – Jun. 2023

Korea Evaluation Institute of Industrial Technology (KEIT) — Materials and Components Technology Development Program

HONORS & AWARDS

The Presidential Science Scholarship — MSIT / Korea Student Aid Foundation (KOSAF)

2022 – 2023

Nationwide selection of 60 third-year undergraduates (2022); full tuition through graduation plus academic stipend. Also selected for the National Scholarship for Science and Engineering (2022), superseded under non-duplication rules.

Outstanding Paper Award — ICROS 2024, Institute of Control, Robotics and Systems

2024

Gold Medal — YU Capstone Design Project Competition, Yeungnam University

2023

Silver Medal — Autonomous Driving Competition, TE Connectivity

2023

Special Award — Autonomous Driving Competition, Yeungnam University	2023
Bronze Medal — Extreme Environmental Drone Competition, Yeungnam University	2023
Silver Medal — Extreme Environmental Drone Competition, Yeungnam University	2022
Grand Award — Self-Following Mobile Robot Competition, Yeungnam University	2021
Silver Medal — Y-Star Startup Item Competition, Yeungnam University	2021

TECHNICAL SKILLS

Software: MATLAB, Python, LabVIEW, SolidWorks, AutoCAD

Hardware & Experiment: Arduino, NI myRIO, 3D Printing, EMG, Metabolic Measurement

LEADERSHIP & SERVICE

President, Yeungnam University Robot Club (Power Supply)	Jan. – Dec. 2023
Republic of Korea Army — Engineer (Military Service)	Apr. 2019 – Nov. 2020